

Customer Sales Analysis Report

Week 5: Advanced Data Manipulation with Pandas

Executive Summary

Metric	Value
Total Revenue	INR 12,365,048
Total Customers	100
Average Order Value	INR 123,650
Top Customer	CUST016 - INR 373,932
Top Product	Laptop - INR 3,889,210
Top Region	North - INR 3,983,635
Overall Churn Rate	10.60%

Dataset Limitations

The sales and churn datasets do not share matching customer IDs, so the merge requirement is demonstrated through a left merge that transparently records zero matched rows. Repeat-purchase retention and basket-level cross-selling are not supported by the supplied sales file because every customer appears once and each row contains one product.

Recommendations

Prioritize Laptop stock and promotions, investigate North region performance, use March as the strongest monthly benchmark, and target month-to-month customers for retention campaigns.